# Finite Semaev row-space closure audit

Independent finite-model validation

26 October 2026

**Status.** This is a scoped computational audit of one frozen Boolean draw. It does not change a Lean theorem, formal target, experiment output, official hypothesis status, or ECDLP cost claim. The executable checker and exact machine-readable output are beside this PDF as `audit.py` and `evidence.json`.

## Question and sources

Does the selected degree-three row span of the fixed $n = 4, n' = 4, s = 1, x_R = 1$ Semaev $S_3$ system satisfy the multiplication closure needed to identify it with a $V_{F,3}$ filtration? The archived equation hash is recorded in `evidence.json`. The repository sources are `research/FFD_SEMAEV_MEASUREMENT1.md`, the corrected replay of 22 October, and the finite Lean row-space certificates. The relevant mathematical boundary is the degree-budget closure definition in Caminata–Gorla, Definition 1.5. Kosters–Yeo, Corollary 4.11 and Remark 4.12 give the trace identity and describe degree-two first falls as usual, rather than universal under every factor-base choice.

## Reproduction and controls

The checker independently evaluates $S_3 = e_2^2 + e_3 + 1$ on all 256 Boolean assignments over $\mathbb{F}_2[X]/(X^4 + X + 1)$, then uses a Möbius transform to recover all four archived component equations. It confirms that $x_R = 1$ is the abscissa of curve points with $y = 6$ and $y = 7$. Boolean products use OR on variable masks and XOR on coefficients. A strict multiplier-degree budget and the historical actual-product-degree row selector yield the same rank and top-degree projection rank for this sample:

| Degree | Strict rows | Product-degree rows | Rank | Projection rank |
|---|---|---|---|---|
| 1 | 1 | 17 | 1 | 1 |
| 2 | 12 | 28 | 11 | 10 |
| 3 | 64 | 80 | 53 | 41 |

The archived 80-draw chart is preserved. The additive chart below labels its 68/12 corrected outputs as *direct-row grades*; their identification with a closed-filtration first-fall count is unverified.

## Closure witness and limit

The published degree-two witness $W$ belongs to the selected rank-53 degree-three span and lies outside the selected degree-two span. Multiplying by $u_0$ gives a degree-three polynomial outside that rank-53 span; adjoining it raises the rank to 54. Since $\deg W = 2$ and $\deg u_0 = 1$, the selected degree-three span is not closed under this Boolean-quotient degree-budget product. This does *not* invalidate the Lean theorem about the literal 80 rows. It shows that a separate bridge is needed before calling that span the full closed $V_{F,3}$ filtration. The rank of the complete closed filtration and the polynomial-ring/Boolean-quotient correspondence are not established here. A reviewed definition and a Lean closure or corrected-row certificate are the next proof obligations.

**Operational limit.** No heavy Lean build was started. GitHub Actions reports a payment-or-spending-limit condition before assigning runners, and the Spot host lacks the required 16 GiB reserve for a heavy local build. The complete report with source links and exact witness masks is in `report.md`.

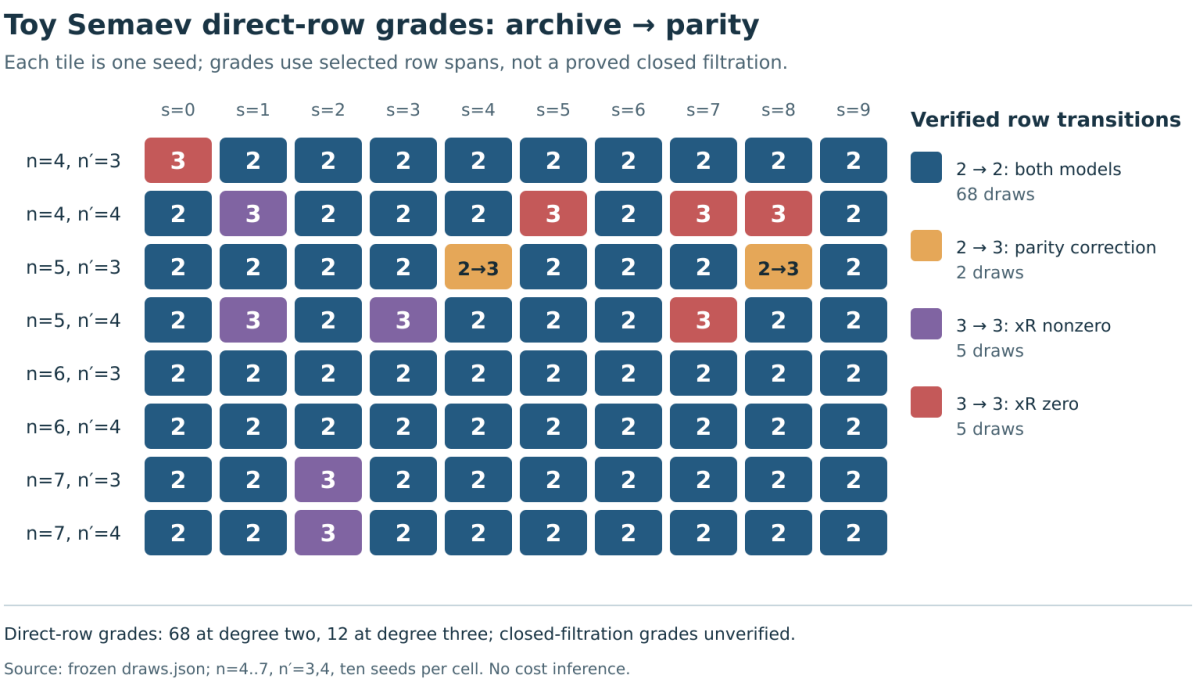



Figure 1: Archived set-product and parity-corrected direct-row grades for eight fixed cells of ten draws. The renderer verifies all 80 tiles against the archived SVG. This is neither a timing plot nor an attack-cost result.